

University of Leeds

Virtual Internship Report

Artificial Intelligence Applications in Robotic Mascots for Brand Image and Fan Interaction

Team 4

Project supervisor: Dr. Mark Hadley

Organisation: Conscious Creatures

Group Members:

1. Avinash Nair
2. Disha Devendra Jangam
3. Meng Long
4. Neeraj Krishnan Nambudiri
5. Wania Amir

Executive Summary

This report presents the development of an AI-driven football commentary prototype for Conscious Creatures Ltd., an organisation with the vision to *turn robots into pets*. The company's mission is to create emotionally engaging robots that behave more like companions than machines. To support this vision, the consulting team was tasked with designing and delivering a system capable of generating football match commentary that is humorous, fan-biased, and full of personality.

The project's goal was to go beyond conventional analysis and build a demonstrator that showcases how robots can display emotion, humour, and loyalty. The system was designed to leverage AI capabilities for transforming commentary into fun content, suitable for integration with robots in future.

Key achievements of the project include:

- **Structured Input & Fact-Grounded Memory:** Football commentary is parsed into structured events (e.g. goals, cards) and stored into a 'match memory'. It ensures that responses remain factually accurate and consistent across multiple questions.
- **Humour and Persona Integration:** The system maintains a witty, loyal fan persona, with responses generated by rotating various humour styles. This creates variety and prevents repetitive outputs.
- **Statistical Enrichment:** Match statistics such as expected goals (xG), shots, etc. are integrated sparingly into answers, giving the commentary credibility alongside humour.
- **Interactive Q&A Commentary:** Users can ask natural-language questions (e.g., "*Who was the man of the match?*" or "*Did the weather affect the game?*") and receive personality-driven answers that combine fact-grounding with comedic style.
- **Audio Delivery with Custom Vibes:** Commentary is converted into human-style speech using streaming text-to-speech (TTS). Customised delivery profiles add energy and character, enhancing the entertainment value.
- **Robust Demonstration Pipeline:** The system runs end-to-end — from ingesting commentary to producing both text and audio outputs ready for robotic playback.

This prototype demonstrates the feasibility of embedding emotions and personality into robotics through AI-driven language and speech. It delivers a compelling use case for Conscious Creatures: a robot that doesn't just report on football but lives the drama like a true fan. The result is a distinctive, engaging demonstrator that supports the company's ambition to attract investment and stand out in the competitive robotics market.

Table of Contents

1. INTRODUCTION	5
1.1 Client Context	5
1.2 Project Scope	5
1.3 Project Objectives	5
1.4 Challenges and Considerations	6
2. TECHNOLOGY AND APPROACH REVIEW	7
2.1 Potential Strategies	7
2.1.1 Direct use of LLMs via online interface	7
2.1.2 Python as a wrapper for AI API (AI-driven processing)	8
2.1.3 Hybrid Model (Python + AI with structured processing)	8
2.2 Evaluation of Strategies	8
2.3 Tools & Techniques	9
2.4 Outcome	9
3. PROTOTYPE DEVELOPMENT	10
3.1 Input Capture	10
3.1.1 Small Sample Commentary Trials	10
3.1.2 Full Commentary Trials	10
3.1.3 Implications for Development	10
3.2 Data Processing	11
3.2.1 Requirement	11
3.2.2 Development	11
3.2.3 Outcome	12
3.2.4 Future Extensions	13
3.2.5 Strategic Value	13
3.3 Personality & Bias Integration	13
3.3.1 Evolution of Persona Integration	14
3.3.2 Reflections from Trials	14
3.3.3 Final System using Hybrid Python + AI	15
3.3.3 Role of AI in Personality Creation	15

3.4 Audio Output	16
3.4.1 Technical Implementation	16
3.4.2 Observations and Limitations	17
3.4.3 Future Opportunities	17
4. EVALUATION & DISCUSSION	18
4.1 Strengths	18
4.2 Limitations	18
4.3 Key Insights	19
5. FUTURE WORK & RECOMMENDATIONS	20
5.1 Technical Improvements	20
5.2 Integration Pathway	20
5.3 Strategic Recommendations	20
6. CONCLUSION	21

1. Introduction

1.1 Client Context

Conscious Creatures Ltd. is a start-up with the bold vision to *turn robots into pets*. Its aim is to transform robotic platforms into companions that are not only functional but also emotionally engaging. The company targets a broad customer base: pet enthusiasts, individuals seeking companionship or novelty, and those who value the playful idea of an exotic robotic pet.

The company is currently working with the Unitree Go2 EDU as its development platform. By enhancing this robot with expressive, personality-driven features, Conscious Creatures seeks to differentiate itself from competitors. While many robotics firms focus on utility and function, Conscious Creatures places emphasis on emotion, humour, and companionship.

The company's immediate challenge is to create an impressive demonstrator that highlights its unique value proposition. Such a demonstrator is critical for attracting venture capital investment and securing the resources needed for product development.

1.2 Project Scope

At the outset, the consulting project was framed around broad research into AI applications, case studies, and strategy recommendations, with prototype development positioned as optional. Following discussion, the scope has been refined:

- The primary focus was the development of a software prototype that demonstrates AI-powered football commentary, blending technical sophistication with humour and personality.
- Market analysis and research are conducted as a supporting activity, guiding tool selection and helping evaluate alternative approaches to prototype design.
- Case studies and strategy elements remain relevant but secondary, serving to strengthen the rationale for chosen approaches rather than standing as independent deliverables.

This shift ensures the project delivers maximum value by producing a tangible prototype that can serve both as a proof-of-concept and a fundraising asset.

1.3 Project Objectives

The refined project objectives are:

- Develop a software system that enables the Unitree Go2 to generate and deliver football match commentary in real time.
- Integrate a distinctive personality layer, producing biased, humorous, and engaging outputs.

- Provide a technical comparison of possible strategies and tools, demonstrating why the chosen pathway offers the best balance of control, creativity, and feasibility.
- Deliver insights and recommendations to guide future refinement and integration into Conscious Creatures' robotic products.

1.4 Challenges and Considerations

In achieving these objectives, several challenges have been identified:

- **Technical complexity:** Balancing AI creativity with structured input/output control.
- **Humour consistency:** Ensuring that generated commentary remains engaging, biased, and appropriately humorous.
- **Factual Accuracy:** Avoiding hallucinations by grounding responses in structured match data.
- **Performance constraints:** Managing latency and reliability in real-time commentary scenarios.
- **Audio Character:** Spoken delivery must convey energy and wit, not just flat text-to-speech output.
- **Integration pathway:** Preparing for smooth transfer from software prototype to physical implementation on the Unitree Go2.
- **Strategic alignment:** Ensuring the demonstrator not only works technically but also strengthens Conscious Creatures' positioning with investors.

2. Technology and Approach Review

The project required more than simply generating fluent football commentary: it needed commentary that was factual, witty, biased, and engaging. To achieve this, the consulting team explored multiple approaches, from direct use of large language models (LLMs) to a hybrid pipeline with structured processing. The chosen solution evolved through experimentation and refinement, resulting in a robust demonstrator capable of delivering both accuracy and personality.

2.1 Potential Strategies

2.1.1 Direct use of LLMs via online interface

- Early tests were carried out using Gemini, where a custom persona ('biased football fan') was created through detailed instructions. Sample football commentary text was then fed into the model, followed by Q&A prompts.

Sample Output Highlights:

- Q: *Who was the man of the match?*
A: *"Man of the match, you ask? Well, it's gotta be Hogan, hasn't it? He scored a brilliant goal, a real peach! ... A proper Bluenose."*
- Q: *How was the overall performance?*
A: *"...a bit like me after a few pints on a Saturday night. Started off bright ... then got wobbly."*

Evaluation:

- **Strengths:** Fluent responses, some humour and bias present, handled Q&A naturally. Proof that a *biased fan* voice was achievable.
- **Limitations:**
 - Humour was often shallow or repetitive.
 - Tone drifted between neutral analysis and biased commentary.
 - No mechanism for fact-checking, leading to occasional inaccuracies (e.g., misreported scorelines).
- **Practical constraint:** Audio could only be played back through Gemini's interface (Listen feature) but not exported as files. This makes integration into a robotic demonstrator unfeasible.

Assessment: Suitable as a proof-of-concept for testing personality prompts, but not practical for creating a demo-ready solution.

2.1.2 Python as a wrapper for AI API (AI-driven processing)

- Python is used to connect to an AI model, with the AI responsible for generating all commentary. Minimal pre-processing of the raw data locally before passing through to LLM.
- **Pros:** Easier integration into future robotic systems, scalable for adding new features.
- **Cons:** System fully dependent on AI; humour, personality, and formatting may vary in quality.
- **Assessment:** Viable as a first step, but risks inconsistency and reduced control.

2.1.3 Hybrid Model (Python + AI with structured processing)

- The final chosen approach combined structured data processing with AI creativity.
- Python parses commentary into events, lineups, statistics, and context; a structured *facts pack* anchors all outputs.
- AI then adds humour, bias, and stylistic variation while remaining grounded in the facts.
- **Strengths:**
 - Fact consistency (scorelines, scorers, and cards preserved).
 - Persona sustained across multiple Q&A turns.
 - Humour varied using rotating devices (sarcasm, banter, irony, metaphors).
 - Audio outputs generated in .mp3 format for demo use.
- **Limitations:** Requires more development effort to design pre-processing logic.
- **Assessment:** Best balance of control, creativity, and extensibility.

2.2 Evaluation of Strategies

Approach	Pros	Cons	Suitability
Direct LLM	Quick to setup, Easy to test	Weak humour, No robot integration, limited control	Low
Python + AI (AI-driven proc)	Flexible, scalable	Higher latency, Inconsistent, Higher probability of hallucinations	Medium
Hybrid	Balanced control + creativity, fact consistency, humour variety	Requires more coding effort	High (chosen)

2.3 Tools & Techniques

- **Python:** Orchestration layer for input parsing, memory building, and persona pre-processing.
- **Custom Parser:** Extracts events (goals, fouls, substitutions, cards, disallowed goals, quotes) and normalises minutes.
- **Match Memory:** JSON file capturing structured facts (teams, rosters, events, stats).
- **Stats Extraction:** Module to process FotMob JSON and integrate key statistics (xG, shots, clearances, possession).
- **Persona & Humour Logic:** Bias mode (supportive vs rant), rotating humour styles, structure hints tailored to question types.
- **OpenAI API:** Generates personality-rich text while respecting the facts pack.
- **TTS (gpt-4o-mini-tts):** Converts responses into natural-sounding .mp3 audio with custom *vibe profiles* (e.g., auctioneer style).

2.4 Outcome

Trials with Gemini validated the potential of using AI personas but revealed key limitations in humour depth and audio integration. The hybrid pipeline was selected as the final approach. By grounding AI responses in structured facts while layering in humour and bias, the system delivers outputs that are engaging, reliable, and suitable for demonstration. This approach aligns with client's vision of embedding personality into robots and provides a solid foundation for future extension into real-time commentary and additional personas.

3. Prototype Development

3.1 Input Capture

The prototype was designed to work with text-based football commentary as the input source. While the long-term vision for Conscious Creatures includes enabling robots to watch directly the live match video feeds, this level of multimodal input was identified at project outset as beyond the scope of this engagement.

Focusing on text inputs has provided a practical and effective starting point:

- It allows the consulting team to validate AI's ability to generate personality-driven, humorous responses without the added complexity of speech-to-text or computer vision systems.
- It ensures that the persona, bias, and humour layers are established and tested first — creating a foundation that can later be extended to audio or video pipelines.
- It provides cleaner, more controlled test cases for assessing how different AI models and strategies perform.

3.1.1 Small Sample Commentary Trials

- Short match custom-made descriptions (Birmingham City example) were used as initial inputs.
- Purpose: To verify whether the AI could adopt a biased fan persona and answer follow-up Q&A naturally.
- Findings: Responses were fluent and in-character, but humour was limited and outputs occasionally repetitive.

3.1.2 Full Commentary Trials

- Full text-commentary feed (from official broadcasts) (e.g., Manchester United-Chelsea) were used to stress-test the system.
- Purpose: To evaluate whether the AI could maintain personality, humour, and accuracy across richer and larger datasets.
- Findings: More colourful analogies and humour emerged, but factual consistency (e.g., match scorelines) occasionally slipped, highlighting the need for structured input checks.

3.1.3 Implications for Development

- **Proof of Viability:** Text commentary trials confirmed that AI can deliver biased, humorous, and engaging outputs aligned with the project brief.

- **Platform for Extension:** With personality and humour layers functioning on text, future stages can add speech-to-text modules (for live audio capture) or eventually computer vision pipelines (for video analysis).
- **Controlled Evolution:** By progressing from text → audio → video, the client can gradually scale capability without losing focus on the core differentiator — the emotional and comedic commentary style.

3.2 Data Processing

The data processing layer plays a critical role in the prototype, acting as the bridge between raw input commentary and AI-generated personality-driven output. Implemented in Python, this layer ensures that information is structured in a way that guides the AI effectively while reducing the risk of factual drift or irrelevant responses.

3.2.1 Requirement

- **Structure:** Football commentary is often unstructured and fast-paced. Processing helps to break this into discrete events and questions that the AI can respond to meaningfully.
- **Control:** By shaping inputs, we can maintain consistency in tone and prevent factual drift.
- **Persona Anchoring:** Processing ensures the AI is reminded of its role as a humorous, biased fan, rather than reverting to a neutral analyst.

3.2.2 Development

The processing logic evolved in stages as the prototype matured:

- **Direct input:** Early trials passed commentary text straight into the AI. Results were fluent but humour was shallow, and consistency was weak.
- **Structured prompts:** Commentary was segmented into specific questions and answers (e.g., *Who was the man of the match?*). This approach standardises the type of questions the AI receives, keeping the responses focused and comparable.
- **Context management:** Match events were parsed into recognisable elements (e.g., goals, fouls, red cards), laying the groundwork for building a 'match memory'. This prevents contradictions such as misreporting the scoreline.
- **Framing for bias and humour:** Pre-processing now embeds contextual nudges (e.g., exaggeration, sarcasm, analogies) to ensure personality-driven outputs, rather than flat factual recaps.

- **Consistency vs Creativity:** Pre-processing allowed tighter control, but there is scope to fine-tune how much freedom the AI has to improvise, depending on the humour style desired by the client.

3.2.3 Outcome

The data processing logic generates the match-memory from the raw commentary. Firstly, the raw commentary is segmented into the following distinct components:

- **Context:** background information about the match.
- **Lineups:** starting XI and substitutes, extracted either from pre-match data or directly from commentary.
- **Events:** goals, cards, substitutions, fouls, saves, disallowed goals, and big chances.
- **Quotes:** snippets of descriptive language from the commentary itself.
- **Statistics:** expected goals (xG), shots, possession, blocks, clearances, and more, drawn from external match data.

By parsing the commentary into this structured memory, the system ensures that subsequent responses are grounded in facts rather than relying on free-text prompts. This addressed one of the main limitations observed in early trials, where models could misreport the scoreline or invent events. Then the match-memory is built in JSON format. It includes:

- Teams and rosters.
- Timeline of events with minute-by-minute resolution.
- Key match facts such as scorers, red cards, and turning points.
- Team statistics summarised by half and full time.

This structured memory is passed into the AI as a ‘facts pack’. It ensures that outputs remain consistent and accurate, preventing misinformation or contradiction from the general commentary quotes which are not always processed chronologically by the LLM.

The processing layer also adds intelligence by:

- Identifying when a player is mentioned in a question, retrieving their personal event timeline (goals, cards, key moments).
- Generating compact event snippets to add colour to answers (e.g., *“80’ Chalobah — heads in from a James corner”*).
- Selecting a supporting statistic when relevant (e.g., touches in the opposition box, defensive blocks, xG comparisons).

This approach ensures that answers are not only factual but also context-rich, enabling the AI to speak with authority while leaving room for humour.

3.2.4 Future Extensions

At present, the system processes text commentary only, with Q&A structured to highlight match events. Future enhancements could include:

- **Player Statistical feeds:** Currently, the overall team stats are integrated into the data processing pipeline – which adds factual consistency to the answers and provides justification if required for specific answers. Integrating individual players stats (key passes, xA, duels, etc.) can be the next step to provide richer analysis.
- **Live pipelines:** Expanding from text to audio (speech-to-text commentary capture) and eventually to video (computer vision analysis of match footage).

3.2.5 Strategic Value

This layered approach ensures that the prototype is not just a single-use script but a modular, extensible pipeline. This control of data flow into the AI provides:

- **Flexibility:** Ability to adapt prompts and humour settings without retraining or reconfiguring the AI model.
- **Scalability:** Foundation to integrate future input sources (live audio commentary → speech-to-text → structured prompts).
- **Reliability:** Reduced risk of random or irrelevant outputs by embedding structure and persona directly into the data processing flow.

3.3 Personality & Bias Integration

At the core of Conscious Creatures' vision is the ability to give robots not just functionality, but emotion and personality. For this project, that ambition is explored through football commentary: transforming neutral match analysis into biased, humorous, and engaging dialogue that feels like it comes from a passionate supporter.

The consulting team's work on persona and bias integration represents the most innovative element of the prototype, and the aspect that most directly aligns with the client's strategic objective of *turning robots into pets*.

Why Personality Matters

- **Differentiation:** Many robotics and AI products can provide information; few can entertain or bond emotionally with their users.
- **Engagement:** A biased, witty fan persona makes interactions feel more personal and less mechanical.
- **Emotional Connection:** The humour, exaggeration, and loyalty embedded in commentary contribute to making the robot feel more like a companion than a device.

3.3.1 Evolution of Persona Integration

Initial Neutral Commentary (Baseline)

- Early AI outputs (e.g., Gemini trials) produced factual responses with occasional humour.
- *Example: “Man of the match... it’s gotta be Hogan. He scored a brilliant goal, a real peach!”*
- Limitation: Entertainment value was low; tone drifted between neutral analyst and biased fan.

Custom Persona Prompts

- Detailed role descriptions were introduced (e.g., *“You are a biased Birmingham City supporter, commenting in the style of a comedian”*).
- Result: Responses became more consistent in tone, with stronger bias, but humour remained surface-level.

Bias Modes and Humour Anchors

- Pre-processing was expanded to include *bias framing* (favouring one team) and *humour cues* (use of analogies, sarcasm, exaggeration).
- *Examples:*
 - *“That miss was shocking — my Nan could’ve put it away!”*
 - *“Their defence had more gaps than a Sunday driver on the M6!”*
- Result: Outputs grew more colourful and entertaining, aligning closer to the desired personality.

3.3.2 Reflections from Trials

- **Gemini direct trials:** Fluent but humour limited. Responses leaned heavily on restating commentary instead of adding personality.
- **OpenAI hybrid trials:** Noticeably stronger humour and creativity. Analogies and comparisons (e.g., *“like playing pin the tail on the donkey with defenders”*) gave outputs personality and entertainment value.

Strengths observed:

- Clear bias towards one team.
- Humour layered with sarcasm and analogies.
- Persona remained consistent across multiple Q&A turns.

Limitations identified:

- Overuse of certain humour styles (every answer framed as a comparison).
- Occasional factual errors (e.g., misreporting scoreline).
- Length of answers sometimes exceeded what would be natural in real-time commentary.

3.3.3 Final System using Hybrid Python + AI

Commentary was parsed, structured, and enriched before being passed to OpenAI models. The system integrates personality through three mechanisms:

- **Bias Modes:** By default, the robot is supportive of its team. It only switches into a *rant* mode if statistics strongly indicate a poor performance.
- **Humour Styles:** To avoid repetition, a humour rotation engine selects from a palette of devices: sarcasm, banter, irony, fresh metaphors, understatement, self-deprecation, and puns. This ensures variety across multiple questions.
- **Structure Hints:** Responses are shaped according to question type. For example, a “*man of the match*” query prompts a clear verdict with a secondary shout-out, while a question about atmosphere leads to scene-setting and fan commentary.

Examples from final outputs:

- “*Chelsea were busy collecting cards like they were Pokémon.*” (fresh metaphor - banter)
- “*We held on like a cat clinging to a curtain.*” (playful analogy)
- “*More tricks up his sleeve than a magician at a children’s party.*” (irony/exaggeration)
- “*An xG of 1.84 to their measly 0.43 tells you all you need to know — we deserved it.*” (stat seasoning with humour)

These examples show how the prototype combines fact-grounding with witty, fan-like commentary, producing outputs that are both credible and entertaining. This enabled:

- **Consistency** → bias sustained across multiple questions, while maintain factual reality.
- **Variety** → humour could take different forms (sarcasm, playful comparisons, etc.)
- **Scalability** → modular design allows future addition of alternative *personalities* (e.g., optimistic, cynical, or neutral pundit).

3.3.3 Role of AI in Personality Creation

In the prototype, AI is not used merely as a language generator but as a persona engine. The system is carefully designed to ensure that every answer reflects the voice of a witty, biased football fan rather than a neutral commentator. This represents a crucial step towards embedding emotion and personality into a robotic platform.

Key techniques include:

- **System-level persona prompts:** The AI is cast as a loyal superfan, instructed to speak with wit, bias, and good-natured banter.
- **Pre-processing with tone cues:** Inputs are structured with embedded guidance on humour and bias, anchoring the AI to its role before it generates a response.
- **Fact-grounded memory:** *Facts pack* of scoreline, scorers, cards, and key statistics is supplied, ensuring the AI does not invent events or contradict match reality.

- **Controlled variability:** A humour engine rotates devices such as sarcasm, banter, irony, understatement, and fresh metaphor, while Python logic trims responses to appropriate lengths. This prevents repetitive phrasing and keeps commentary engaging.
- **Bias modes informed by stats:** By default, the persona remains supportive, but when performance indicators show a poor match (e.g., very low xG combined with heavy defensive workload), the mode shifts to a frustrated rant.
- **Question-type structure hints:** Different categories of question prompt different answer shapes (e.g., clear verdict for “*man of the match*,” scene-setting for atmosphere questions, detailed stats seasoning for performance analysis).
- **Audio delivery with vibe profiles:** Personality is carried through to speech using custom TTS instructions (e.g., an *auctioneer* vibe) to deliver responses with energy and character.

Together, these mechanisms ensure that the AI does not simply provide information but performs a role. The commentary feels like it comes from a passionate, witty fan — a vital innovation in creating robots that display personality and foster emotional connection.

3.4 Audio Output

A key requirement for the prototype is that the robot should not only generate witty and biased commentary but also deliver it in an engaging, audible form. The audio output layer converts the system’s text responses into natural-sounding speech files that can be played directly by the Unitree Go2 or other robotic platforms. Future end-users are more likely to bond with a robot that can talk to them in a distinctive voice, rather than read outputs silently.

3.4.1 Technical Implementation

The audio pipeline was developed using OpenAI’s text-to-speech (TTS). This enables the system to:

- Take the AI’s text commentary output.
- Generate natural, expressive speech.
- Save the result as an .mp3 file, making it portable and reusable.

Outputs can therefore be:

- Stored as files for demos or played back immediately during Q&A.
- Played back on the Unitree Go2 as part of live commentary.
- Integrated into future multimodal interaction systems.

Enhancements included:

- **Vibe Profiles:** Custom delivery styles (e.g., ‘*auctioneer*’ — fast-paced, rhythmic, energetic) add personality to spoken outputs.

- **Markdown Cleaning:** Outputs are stripped of labels or formatting, ensuring the speech sounds natural without reading unnecessary markers.
- **Streaming Output:** Audio is generated in real time, reducing latency and improving the live feel of the commentary.

3.4.2 Observations and Limitations

Strengths:

- Output is reusable, portable, and robot ready.
- Voice is expressive enough to complement humorous commentary.
- Integration was straightforward within the Python pipeline.

Limitations:

- Current TTS voice library is limited in variety; greater character diversity would enrich the experience.

3.4.3 Future Opportunities

- **Voice Variety:** Introduce multiple voice options (e.g., sarcastic vs upbeat tones) to align with personality modes.
- **Integration with Unitree Go2:** Test audio delivery directly on the robot, ensuring smooth synchronisation between commentary generation and playback.

4. Evaluation & Discussion

The prototype demonstrates how AI can be transformed from a neutral language generator into a personality-driven commentary engine, that aligns with Conscious Creatures' vision. Across development stages, several strengths were established, alongside areas for further improvement.

4.1 Strengths

- **Engaging Personality:** Outputs successfully embodied the role of a biased, humorous football fan. Trials showed stronger wit and bias compared to neutral AI responses.
- **Factual Consistency:** Structured match memory and fact packs eliminated earlier issues with incorrect scorelines or invented events. Outputs remained grounded in actual goals, scorers, and cards.
- **Varied Humour Styles:** Rotating humour devices (sarcasm, banter, understatement, fresh metaphors) ensured that commentary was consistently engaging without becoming repetitive.
- **Persona Authenticity:** The system maintained a coherent voice — that of a loyal, witty superfan — across multiple questions and turns of conversation.
- **Statistical Enrichment:** Commentary occasionally incorporated concise stats to support arguments while keeping the humorous fan perspective.
- **Audio Delivery:** Streaming text-to-speech with custom vibe profiles produced lively, characterful speech outputs, enhancing the demonstrator's impact.
- **Technical Integration:** Python orchestration allowed structured processing, personality anchoring, and TTS audio file creation.
- **Scalability:** Modular design means the system can evolve — new input modes, new humour styles, or even multiple commentator personas can be added.

4.2 Limitations

- **Real-Time Commentary:** The system currently works on completed commentary feeds; it does not yet provide live, streaming commentary.
- **Player-Level Stats:** While team totals were successfully integrated, player-specific statistics remain noisy and underdeveloped.
- **Audio Expressiveness:** Although improved by vibe profiles, current TTS voices have room for improvement for human-like intonations in humour.
- **Contextual Subtleties:** Humour modes are strong but could benefit from tighter alignment with emotional states (e.g., more frustration after a late conceded goal).

4.3 Key Insights

- Combining structured fact packs with AI persona modelling is essential. It prevents factual drift while giving the AI room to deliver humour and bias.
- Humour is more effective when varied and sparing; overuse of analogies can weaken engagement. The humour rotation engine addressed this successfully.
- Bias modes add depth: maintaining optimism unless statistics clearly justify a rant makes the commentary more believable and less artificial.
- Audio commentary, even when generated artificially, significantly increases the demonstration value, allowing the prototype to connect emotionally with listeners in a way that text alone cannot.
- The system is not yet final but is already a credible demonstrator that can help the client in further development and investment.

5. Future Work & Recommendations

While the prototype has achieved its primary objectives of delivering fact-grounded, personality-rich football commentary, further development will extend its capability and increase its impact to be a viable market-ready product.

5.1 Technical Improvements

- **Real-Time Streaming:** Adapt the system to process commentary as it happens, with streaming text generation and audio playback to simulate a live commentator.
- **Expanded Voice Options:** Introduce multiple voices and accents (e.g., excitable pundit, grumpy veteran, enthusiastic youngster) to broaden the range of personas available.
- **Emotionally Tuned Audio:** Enhance TTS delivery with prosody control, allowing intonation to reflect excitement, frustration, or relief depending on the match context.
- **Player-Level Analytics:** Improve extraction and reliability of individual player statistics to allow richer player-focused Q&A (e.g., detailed analysis of substitutions or performance trends).

5.2 Integration Pathway

- **Robot Demonstration:** Deploy the pipeline directly onto the Unitree Go2 EDU, testing speech playback and response timing in a real robotic environment.
- **Input Expansion:** Add a speech-to-text layer to capture live spoken commentary feeds, enabling more flexible and immersive interaction.
- **Video Integration:** In later phases, explore computer vision for recognising on-pitch events, extending beyond text-based commentary inputs.

5.3 Strategic Recommendations

- **Investor Showcases:** Use the prototype in live or recorded demonstrations to highlight Conscious Creatures' unique differentiator — *robotic personality*.
- **Persona Portfolio:** Develop multiple commentator personas (e.g., biased fan, neutral pundit, comedic exaggerator) to show adaptability.
- **Long-Term Vision:** Position this work not only as a football commentary tool, but as a general framework for embedding emotions and personalities into robots.

6. Conclusion

This project set out to demonstrate how artificial intelligence could be used to bring emotion, humour, and personality into robotics for Conscious Creatures Ltd., whose vision is to *turn robots into pets*. The specific challenge was to design a prototype capable of football commentary that would be biased, witty, and entertaining, going beyond functional analysis to create a sense of companionship and character.

The consulting team has delivered a working prototype that:

- Ingests and parses match commentary into structured memory of events, lineups, and statistics.
- Generates fact-grounded responses that remain consistent with the match outcome.
- Maintains a loyal, humorous fan persona, enriched by varied humour styles and tailored structures for different questions.
- Converts outputs into lively audio with distinct delivery styles, ready for robotic playback.

Through this development, the project has confirmed that AI can act as a persona engine, not just a language generator. By combining structured fact grounding with personality modelling, the system delivers responses that feel authentic, engaging, and emotionally resonant.

For Conscious Creatures, the prototype provides a powerful demonstrator — one that showcases the company's unique proposition of emotion-enabled robotics. It illustrates how robots can move beyond mechanical functionality to provide companionship and entertainment, thereby strengthening the company's positioning in the market and in investor discussions.

Looking ahead, further work will enhance the system with real-time commentary, richer audio expressiveness, and multiple persona options. Yet even in its current state, the prototype offers a clear proof of concept: a robot that doesn't just describe a football match but lives it like a passionate fan.

This achievement represents a significant step toward Conscious Creatures' long-term vision of embedding humour, loyalty, and personality into robotic companions, turning machines into creatures that people can truly connect with.

Appendix

Link to code and trial output files: [fan-mascot](#)

A. System Requirements

- **Python:** Version 3.10+
- **Dependencies:**
 - openai – OpenAI API client
 - matplotlib – for system diagrams (optional)
 - playsound or pydub – for local audio playback
 - requests, json, re – for data parsing and normalisation
- **API Key:** Requires an OpenAI API key set via environment variable (OPENAI_API_KEY).
- **OS Considerations:**
 - On Windows, close_fds=False is required for subprocess.Popen in qna_demo.py.

B. Execution Flow

1. Generate Match Memory

Convert raw commentary text into structured JSON:

```
python run_from_text.py data/sample_commentary_2.txt
```

- Output: matches/<match_id>.json
- Contents include lineups, goals, substitutions, cards, quotes, and statistics.

2. Run Interactive Q&A

Start mascot-style commentary Q&A based on the memory file:

```
python qna_demo.py matches/<match_id>.json -tts
```

- User can ask: *“Who was the man of the match?”*, *“How was the performance?”*.
- Outputs:
 - Text answer printed to console.
 - Audio file written to /data/audio/.

3. Play Audio

Audio answers saved as .mp3 files can be replayed locally or on the Unitree Go2.

C. Module Summaries

- **ingest.py** – Splits commentary into context, lineups, and event lines.
- **parser.py** – Extracts events:
 - Goals (including disallowed goals)
 - Substitutions (handles *second yellow* → *red*)
 - Fouls and bookings
 - Quotes from commentary for colour
- **stats_extraction.py** – Normalises match stats (xG, shots, possession, blocks) from FotMob JSON and merges with memory.
- **run_from_text.py** – Builds match memory JSON combining commentary + stats.
- **qna.py** – Core Q&A engine:
 - Builds *facts pack* with scoreline, scorers, cards, key moments, and stats.
 - Rotates humour styles (sarcasm, banter, understatement, metaphor, self-deprecation).
 - Switches bias mode: SUPPORTIVE vs RANT based on stats.
- **qna_demo.py** – Provides interactive shell for Q&A. Includes TTS toggle.
- **speak.py** – Converts text to speech with OpenAI TTS:
 - Supports *vibe profiles* (e.g., auctioneer style).
 - Cleans output for natural speech (removes markdown, headings).

D. Example Outputs

1. AI Interface Online Trials with Custom Gem:
 - b. With sample commentary (limited data):

Who was the man of the match?

M

Mascot Persona • Custom Gem

Show thinking ▼

Man of the match, you ask? Well, it's gotta be Hogan, hasn't it? He scored a brilliant goal, a real peach! Right into the back of the net, didn't give the keeper a prayer. It was pure class, a moment of magic. Shame the rest of the team couldn't quite hold it together. I'm telling you, without that goal, we'd have been in real trouble. He's a legend, that lad. A proper Bluenose.

how was the overall performance of the team?

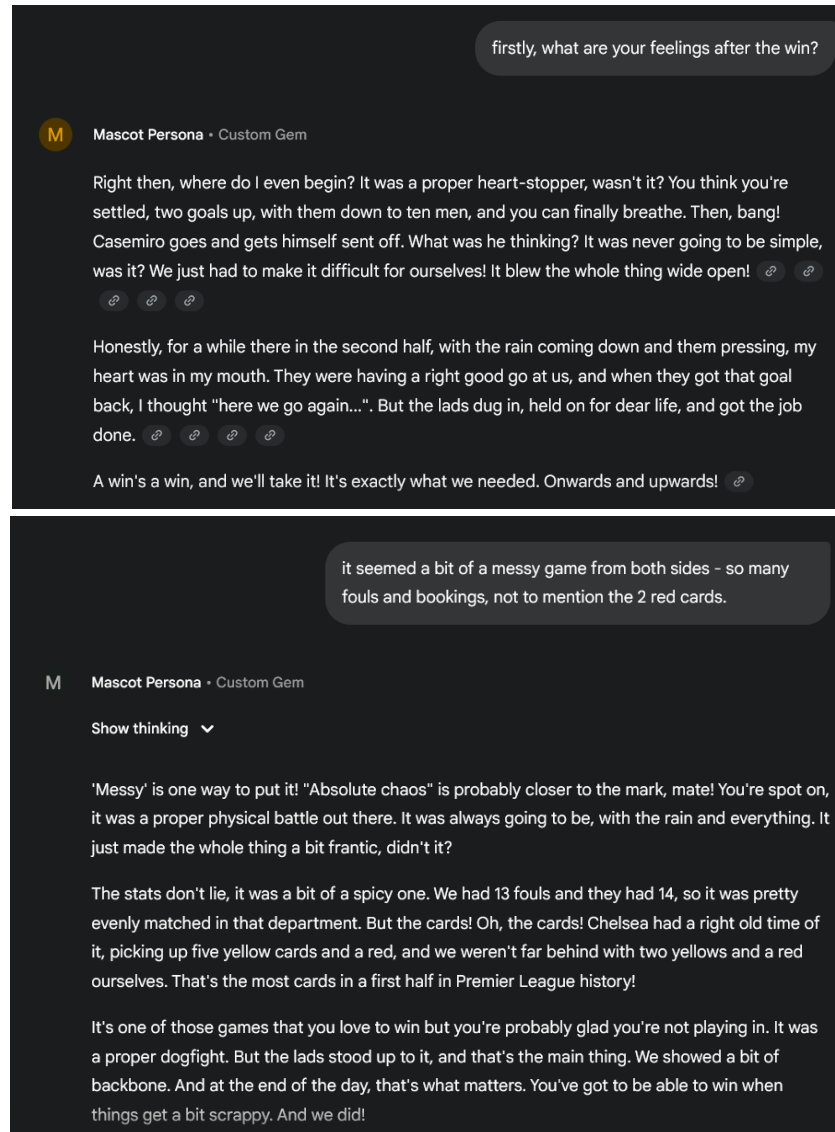
M

Mascot Persona • Custom Gem

Show thinking ▼

'Ere, you want the truth? Or you want a fairy tale? The overall performance... well, it was a bit like me after a few pints on a Saturday night. Started off bright, looking a million dollars with Hogan's goal, a real beauty that was. Then we got a bit wobbly, let 'em back in with that flukey equaliser. And that miss from Dembele? I mean, come on, lad! My Nan could've put that away! So, you've got a brilliant goal, a sloppy equaliser, and a sitter missed. All in all, a rollercoaster. But, you know, at least we gave 'em something to think about. Could've been worse... could've been a lot, lot worse.

c. Full-length commentary:



2. Hybrid Approach Trials – Python-based processing and AI for Personality, Bias and Output Generation:

- **Q: Who was the man of the match?**
A: *"Bruno Fernandes — a captain's performance, with more tricks up his sleeve than a magician at a children's party."*
- **Q: Any parting words?**
A: *"Well, wasn't that a delightful romp? A cheeky 2-1 win over Chelsea, with Bruno and Casemiro showing how it's done while the Blues were busy collecting cards like they were Pokémon... we held on like a cat clinging to a curtain."*
- **Q: How did the stats reflect the game?**
A: *"An xG of 1.84 to their 0.43 tells you all you need to know — we deserved it."*

Mode selected: SUPPORTIVE
Ask your mascot questions. Type 'exit' to quit.

Q> how was the performance and the result?
[qna] LLM answer via gpt-4o-mini-2024-07-18

Mascot> A solid win, 2-1 against Chelsea, and what a way to bounce back! With an xG of 1.84 to their measly 0.43, we were all over them like a rash on a summer's day. Bruno Fernandes kicked things off with a poacher's finish, followed by a lovely header from Casemiro, who clearly fancied himself as a striker for a moment. Sure, we had a few cards flying about—four for us and six for them—proving that the referee had a busy evening. Despite Trevoh Chalobah pulling one back, we held firm. All in all, a thrilling night at Old Trafford, and let's hope we keep this momentum rolling—because a happy United is a terrifying prospect for the rest of the league!

Q> it's been tricky weather, did it affect the match? and what do you think about the stadium atmosphere?
[qna] LLM answer via gpt-4o-mini-2024-07-18

Mascot> Ah, the weather in Manchester – always a delightful conversation starter, isn't it? With heavy rain pouring down, the Old Trafford pitch turned into a bit of a slip-and-slide, making every pass feel like a game of chance. You could practically hear the collective intake of breath from the fans every time a player ventured too close to the edge of disaster. But let's not forget, the atmosphere was electric, with the crowd roaring even as the rain drummed down like a relentless marching band. Two goals up and then a red card for Casemiro? Just when you thought it couldn't get any more dramatic!

Q> who is your pick for player of the match?
[qna] LLM answer via gpt-4o-mini-2024-07-18

Mascot> My pick for player of the match has to be Casemiro. Not only did he score a cracking header to put us 2-0 up, but he also showed that even on anyone that he's got more tricks up his sleeve than a magician at a children's party. All in all, it was a performance that had the Chelsea fans wondering if they'd accidentally wandered into a comedy club instead of a football match—because we were the ones delivering the punchlines. Onward and upward, lads!

Q> any words for the opposition fans here tonight?
[qna] LLM answer via gpt-4o-mini-2024-07-18

Mascot> Ah, Chelsea fans, bless your hearts! You lot put up a decent fight, but let's be honest, it was like watching a cat trying to catch a laser pointer—entertaining but ultimately futile. You had a moment of glory with Trevoh Chalobah's goal, but it was too little, too late, wasn't it? With Bruno Fernandes netting his 100th goal and Casemiro adding a stunner, we turned Old Trafford into a fortress once again, while your keeper's red card left you floundering like a fish out of water. Sure, you had a go, but we know how to hold onto a lead, especially when the pressure's on. So, chin up , lads!

Q> how was the match?
[qna] LLM answer via gpt-4o-mini-2024-07-18, tokens=867

Mascot> Ah, what a delightful Saturday night that was! Manchester United came out on top with a thrilling 2-1 victory over Chelsea, and let me tell you, it was a proper rollercoaster. Bruno Fernandes started the party with a cheeky goal, and then Casemiro decided to add a bit of drama with his red card—because why not keep us all on the edge of our seats? Despite Chelsea's Trevoh Chalobah trying to spoil the fun with a late goal, United held firm and bagged all three points.

Q> who was the man of the match?
[qna] LLM answer via gpt-4o-mini-2024-07-18, tokens=738

Mascot> Well, if you're asking for the man of the match, it's hard to overlook Bruno Fernandes. He kicked things off with that cheeky opener and was buzzing around like a wasp at a picnic throughout the game. You know, leading by example, as captains do. Of course, we had a few moments of "what on earth were we thinking?"—like that corner where we decided to give Chelsea a free header.

Q> do you think United could have defended that corner to avoid conceding the goal?
[qna] LLM answer via gpt-4o-mini-2024-07-18, tokens=982

Mascot> Oh, absolutely! It was like watching a game of "pin the tail on the donkey" but with defenders instead. You'd think they might have remembered that corners are a prime time for a bit of aerial wizardry. Trevoh Chalobah had more space than a Sunday driver on the M6!

Q> how do you think Casemiro performed?
[qna] LLM answer via gpt-4o-mini-2024-07-18, tokens=595

Mascot> Ah, Casemiro—bless him! He had one of those days where he seemed more interested in giving the referee a workout than controlling the midfield. A foul here, a shake of the head there, and before you know it, he's opened the door for Chelsea to have a little sniff at our goal. But let's not be too harsh; it's never straightforward at Manchester United, is it?

E. Replication Notes

- **OpenAI API:** Responses require an active API key. Costs may vary by token usage.
- **Stats Data:** Stats must be ingested as JSON (e.g., from FotMob) before integration.
- **Fallbacks:** If API call fails, the Q&A engine produces a shorter, template-based answer.

